How to Use This SQL Guide

SQL is central to the BI3 technical round - the 'second highest salary' query was explicitly reported.

Each problem gives the schema with sample data, the question, a verified query, expected output and an explanation. Every query was executed in SQLite.

SQL Problems

Problem 1: Second Highest Salary

Difficulty: Easy | Pattern: Subquery | Authenticity: Verified Previous-Year

Schema & Sample Data

```
CREATE TABLE employee (id INTEGER, name TEXT, salary INTEGER);
INSERT INTO employee VALUES (1,'Kiran',30000),(2,'Divya',55000),(3,'Arun',55000),(4,'Bala',4200
```

Question

Find the second highest distinct salary. (Explicitly reported in BI3 technical round.)

Solution Query

```
SELECT MAX(salary) AS second_highest
FROM employee
WHERE salary < (SELECT MAX(salary) FROM employee);
```

Expected Output

```
second_highest
42000
```

Explanation

Inner query finds top salary 55000; outer takes the max below it (42000), skipping duplicate tops.

Problem 2: Nth Highest Salary (LIMIT/OFFSET)

Difficulty: Easy-Medium | Pattern: ORDER BY + LIMIT | Authenticity: Community Reported

Schema & Sample Data

```
CREATE TABLE emp (id INTEGER, salary INTEGER);
INSERT INTO emp VALUES (1,100),(2,200),(3,200),(4,300),(5,400);
```

Question

Find the 3rd highest distinct salary using ORDER BY with LIMIT/OFFSET.

Solution Query

```
SELECT DISTINCT salary
FROM emp
ORDER BY salary DESC
LIMIT 1 OFFSET 2;
```

Expected Output

```
salary
200
```

Explanation

Distinct desc: 400,300,200,100. OFFSET 2 skips 400,300; LIMIT 1 returns 200.

Problem 3: Total Sales per Region

Difficulty: Easy | Pattern: GROUP BY / SUM | Authenticity: Community Reported

Schema & Sample Data

```
CREATE TABLE sales (region TEXT, amount INTEGER);
INSERT INTO sales VALUES ('North',100),('South',200),('North',150),('East',50),('South',100);
```

Question

Total sales per region, ordered by total descending.

Solution Query

```
SELECT region, SUM(amount) AS total_sales
FROM sales
GROUP BY region
ORDER BY total_sales DESC;
```

Expected Output

```
region|total_sales
South|300
North|250
East|50
```

Explanation

GROUP BY region + SUM aggregates each region: South 300, North 250, East 50.

Problem 4: Join Orders with Customers

Difficulty: Easy-Medium | Pattern: INNER JOIN + GROUP BY | Authenticity: Community Reported

Schema & Sample Data

```
CREATE TABLE customers (id INTEGER, name TEXT);
CREATE TABLE orders (oid INTEGER, cid INTEGER, amount INTEGER);
INSERT INTO customers VALUES (1,'Ram'),(2,'Sita'),(3,'Lax');
INSERT INTO orders VALUES (10,1,500),(11,1,300),(12,2,700);
```

Question

Total ordered amount per customer who has orders, ordered by name.

Solution Query

```
SELECT c.name, SUM(o.amount) AS total
FROM customers c
JOIN orders o ON c.id = o.cid
GROUP BY c.name
ORDER BY c.name;
```

Expected Output

```
name|total
Ram|800
Sita|700
```

Explanation

INNER JOIN drops Lax (no orders); Ram 500+300=800, Sita 700.

Problem 5: Departments With More Than 2 Employees

Difficulty: Medium | Pattern: GROUP BY + HAVING | Authenticity: Community Reported

Schema & Sample Data

```
CREATE TABLE emp (id INTEGER, dept TEXT);
INSERT INTO emp VALUES (1,'A'),(2,'A'),(3,'A'),(4,'B'),(5,'B'),(6,'C');
```

Question

List departments with more than 2 employees and their counts, ordered by department.

Solution Query

```
SELECT dept, COUNT(*) AS headcount
FROM emp
GROUP BY dept
HAVING COUNT(*) > 2
ORDER BY dept;
```

Expected Output

```
dept|headcount
A|3
```

Explanation

HAVING filters groups after aggregation; only A (3) exceeds 2.

Problem 6: Monthly Order Totals

Difficulty: Medium | Pattern: GROUP BY derived column | Authenticity: Research-Backed Company Style

Schema & Sample Data

```
CREATE TABLE orders (id INTEGER, order_date TEXT, amount INTEGER);
INSERT INTO orders VALUES (1,'2024-01-05',100),(2,'2024-01-20',200),(3,'2024-02-10',150),(4,'20
```

Question

Total order amount per month (YYYY-MM), ordered by month.

Solution Query

```
SELECT substr(order_date, 1, 7) AS month,
       SUM(amount) AS total
FROM orders
GROUP BY month
ORDER BY month;
```

Expected Output

```
month|total
2024-01|300
2024-02|300
2024-03|300
```

Explanation

substr extracts YYYY-MM as the grouping key; each month sums to 300.

Problem 7: Highest Paid Employee per Department

Difficulty: Hard | Pattern: Window (RANK/PARTITION) | Authenticity: Research-Backed Company Style

Schema & Sample Data

```
CREATE TABLE staff (name TEXT, dept TEXT, salary INTEGER);
INSERT INTO staff VALUES ('A','Sales',40000),('B','Sales',60000),('C','Tech',90000),('D','Tech'
```

Question

Return the highest-paid employee in each department, ordered by department.

Solution Query

```
SELECT dept, name, salary
FROM (
  SELECT dept, name, salary,
         RANK() OVER (PARTITION BY dept ORDER BY salary DESC) AS rnk
  FROM staff
) t
WHERE rnk = 1
ORDER BY dept;
```

Expected Output

```
dept|name|salary
Sales|B|60000
Tech|C|90000
```

Explanation

RANK() partitioned by dept ranks salaries; keep rank 1: Sales B 60000, Tech C 90000.

BI3 Technologies

SQL Problems

Applied Role(s):

Data Engineer - Trainee

Online Assessment Preparation Material

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Every question carries an authenticity label. See the legend on the next page.

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Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

Label	Meaning
Verified Previous-Year	The specific problem/pattern is documented from actual reported assessment experiences.
Frequently Repeated	The topic or pattern type is documented as recurring across multiple hiring cycles.
Community Reported	A widely shared, role-relevant pattern reported across placement communities and forums.
Research-Backed Company Style	Authored to match the researched difficulty, style and pattern - practice, not a real past question.

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.